



# Industrial furnaces with thermochemical waste-heat recuperation by coal gasification

Dmitry Pashchenko

Samara State Technical University, 244 Molodogvardeiskaya str., Samara, 443100, Russia



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## ABSTRACT

A thermochemical waste-heat recuperation (TCR) system by coal gasification is considered. Two concepts of TCR were analyzed: gasification with steam and gasification with the steam-flue gas mixture. To determine the effectiveness of using TCR systems by coal gasification, a thermodynamic analysis of the gasification process was carried out in the wide ranges of the operation parameters: a temperature range is 200–1200 °C, the steam-to-carbon (S/C) ratio is 0.25–2.0. The thermodynamic analysis has been realized via Aspen HYSYS software. To evaluate the effectiveness of the thermochemical recuperation systems by coal gasification, the heat recuperation degree (HRD) was used as an efficiency criterion of such systems. The algorithm of the calculation of HRD is proposed. It was found in the temperature range up to 700–800 °C HRD increases monotonically and reaches its maximum value at 720 °C for S/C = 1.0, 810 °C at S/C = 0.5 and 930 °C at S/C = 0.25. There is a heat deficit when S/C above 1.0 in the inlet reaction mixture. Thus, two types of HRD were introduced – gross HRD and net HRD. The gross HRD shows the relation between the total recovered heat and heat of exhaust gases. The net HRD shows the relation between the total recovered heat without the enthalpy of residual steam and heat of exhaust gases. The maximum net HRD of 0.97 can be achieved at S/C = 1.0 at a temperature of 800 °C. The heat transformation coefficient (HTC) was analyzed. An increase in the S/C ratio leads to an increase in HTC. With an increase in the S/C ratio above the stoichiometric ratio, HTC increases slightly. When the S/C ratio increases, the specific low heating value of synthesis gas (per 1 kg of syngas) decreases.

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## 1. Introduction

Coal is one of the oldest types of fuels used to produce various types of energy: thermal, electrical, mechanical work, etc [1]. In addition, coal is widely used as a technological raw material for the production of various chemical products (mothballs [2], phenol [3], pitch [4]). Currently, despite the active development of renewable energy technologies as well as falling prices for oil [5] and natural gas, coal is widely used in metallurgy and energy [6,7]. The share of coal in the fuel and energy balance of different countries varies from 0% (in some countries of the European Union) to more than 60% in China [8,9]. Given that approximately 80% of China's carbon dioxide emissions come from burning coal, coal combustion in China produced more than 20% of global total energy-related carbon emissions in 2017 [10]. The main disadvantage of coal as a fuel is a harmful environmental impact [11]. That is why some countries have abandoned or plan to abandon the use of coal as

fuel. However, a large share of coal in the global energy balance requires the search for promising solutions for using coal as a fuel.

Coal is widely used in metallurgy. It is commonly known that metallurgical furnaces have low energy efficiency due to large losses of heat with exhaust gases [12,13]. In some cases, the temperature of the exhaust gases after the furnace can reach up to 1000–1100 °C. In order to increase the energy efficiency of such furnaces, recuperation of the exhaust heat is used. Various recovery methods can be used: air heating air and/or fuel before combustion [14] or thermochemical recuperation. Thermochemical recuperation (TCR) is a promising way for increasing the energy efficiency of various fuel-consuming plants including the industrial furnaces. The main idea of thermochemical waste-heat recuperation is the use of the waste-heat to produce a new fuel by endothermic transformation of an initial fuel [15]. In this case, the new endothermally processed fuel has a combustion heat (in terms of unit mass of the original fuel) than a combustion heat of the original fuel. This new fuel is sometimes called synthetic fuel or synfuel [16]. The synfuel gas is used either partially or completely as fuel in

E-mail address: [pashchenkodmitry@mail.ru](mailto:pashchenkodmitry@mail.ru).

such a fuel-consuming plants with TCR [17]. In the TCR systems, there is a transformation of the enthalpy of flue gases into the chemical energy of synfuel.

In recent years, many works on thermochemical recuperation in various fuel-consuming equipment have been published: industrial furnaces [13], gas turbine units [18], internal combustion engines [19–21], etc. Any organic fuel, the oxidation of which gives an endothermic effect, can be used for thermochemical recuperation. The main criteria for the choosing of the endothermic process for TCR are presented in the author's previous work [22]. According to the literature review, the strong interest of researchers is directed to the use of natural gas [23], liquid hydrocarbons [24,25], and alcohols [26,27] as fuel for TCR.

The results of an investigation of the advances natural gas-fired turbine systems with thermochemical recuperation of exhaust heat were presented by Rabovitser et al. [28]. The authors presented the supplementary opportunity for energy efficiency as well as environmental performance of the gas turbines when thermochemical recuperation is used. In the mentioned paper, the methane is considered as an original fuel and steam methane reforming is used as an endothermic process. The authors reported that system efficiency of the natural gas-fired turbine systems was achieved up to 68%. Moreover, they presented that  $\text{NO}_x$  emission less than 6 ppm may be obtained by combustion of new synfuel produced the TCR system.

Popov et al. considered the TCR system by methane reforming in a glass melting furnace [12]. The authors considered as an oxidizer of methane three substances: steam, flue gases, and steam-flue gases mixture. They paid attention that TCR leads to significant increasing in energy efficiency (up to 16–25%) as well as decreasing in  $\text{NO}_x$  emission when synfuel is used as fuel in the furnace. A series of articles on thermochemical recuperation by methane reforming was published by Gaber and co-authors [17,29]. In these articles, the authors reported the benefits of using the TCR systems by natural gas reforming such as energy efficiency and environmental performance of fuel-consuming equipment.

Chakravarthy et al. presented the study of the theoretical potential of thermochemical exhaust heat recuperation for internal combustion engines (ICE) [19]. The results of second law analysis for various liquid fuels such as iso-octane, ethanol and methanol were presented. The authors reported that use of TCR for ICE leads to considerable increase in second law efficiency for all investigated fuels: 3–5% for methanol, 9% for ethanol, 11% for iso-octane. Moreover, they noted that reduced combustion irreversibility when synfuel is used as fuel in ICE can be considered as benefits for future TCR implementations.

In author's previous paper, the comparison study of thermochemical waste-heat recuperation by steam reforming of liquid biofuels was performed [30]. Four type of liquid biofuels were tested: methanol, ethanol, butanol, and glycerol. As it was shown, that maximum efficiency of the use of TCR is at flue gas temperature of 600 K for methanol, 700 K for ethanol, 850 K for butanol, 900 K for glycerol. In addition, Gaber et al. presented the results of thermochemical analysis and experimental investigation of a recuperative waste heat recovery system for the tri-reforming of light oil [24]. In this paper, the authors investigated various operating conditions for the TCR systems.

The type of fuel for TCR depends on the operating conditions, in particular, temperature. Natural gas in the TCR system is used for the high-temperature flue gases and liquid fuel is used for the moderate low temperature of flue gases. If we draw a parallel between hydrocarbon fuel and coal, it will become obvious that coal can be used for thermochemical recuperation as well. The main condition for this is the implementation of the endothermic transformation of coal into synthesis gas. Such an endothermic

transformation is a well-studied process of coal gasification.

Kuchonthara et al. based on the concept of thermochemical recuperative energy proposed an integrated power generation cycle combining thermochemical recuperation, brown coal gasification, and a solid oxide fuel cell (SOFC) [31]. The authors suggested using the exhaust heat for coal gasification. The thermodynamic analysis was performed via Aspen Plus program. The modeling results depicted that the cycle efficiency increases by up to 39.5% (HHV) with TCR. In addition, Hayashi and co-authors reported the results of the investigation of thermochemical energy recuperation for hydrogen production and power generation by gasification of solid fuels [32].

The gasification process is endothermic and its implementation requires the supply of external heat. In thermochemical recuperation systems, such an external heat source for endothermic processing of the original fuel is exhaust heat. Therefore, the aim of the work is to evaluate the energy efficiency of coal use in the TCR systems for various configurations, as well as to determine the effect of operating parameters, such as temperature, pressure, and composition of the initial reaction mixture, on the efficiency of the TCR systems by coal gasification.

## 2. Methodology

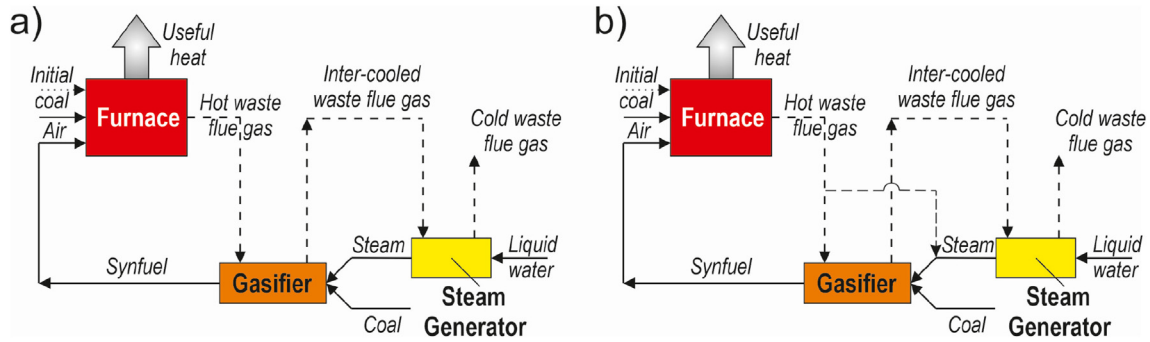
### 2.1. TCR concept

The main concept of thermochemical waste-heat recuperation is the transformation of waste-heat into the chemical energy of new synthetic fuel (synfuel). The schematic diagram of an industrial furnace with thermochemical recuperation of the waste heat by coal gasification is presented in Fig. 1. Fig. 1 contains two schematic diagrams: a) TCR by coal gasification with steam; b) TCR by coal gasification with steam-flue gases mixture. In these schematic diagrams, the flue gases after the furnace are passing through a gasifier and then through a steam generator. In the gasifier, the coal gasification process with external heat from flue gases is occurring. The steam is generated in the steam generator by the residual heat of flue gases. In the scheme in Fig. 1b the flue gases after a furnace is divided into two streams by a gate. The first stream is used to supply external heat for the gasification process. The second stream is directed to a reaction space of the gasifier. For this scheme, the carbon dioxide from flue gases is used as an oxidizer of coal.

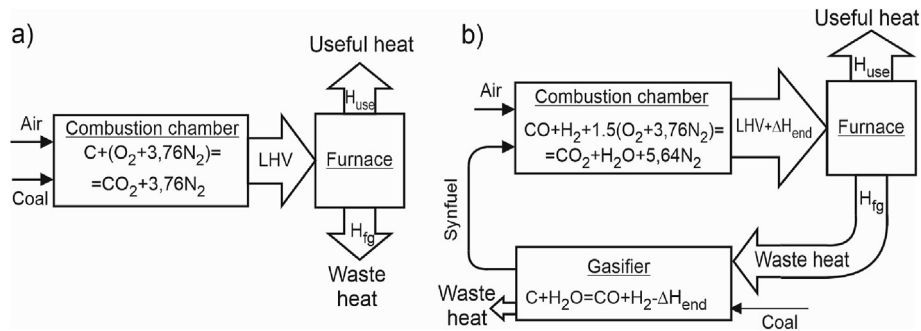
In traditional coal combustion furnaces, coal is burned directly in the combustion chamber. In the furnaces with TCR by coal gasification, coal is initially gasified to synfuel in the gasifier. therefore, the heat generation in the furnaces with TCR is divided into two steps. The first stage is coal gasification to increasing of LHV. The second stage is the combustion of new synthetic fuel that has higher LHV than that of the original coal ( $\text{LHV} + \delta H_{\text{end}}$ ). The comparison of the heat generation processes for the traditional furnaces (a) and the furnaces with TCR (b) are depicted in Fig. 2.

In the gasifier, various chemical reactions are taking place. The main reactions in the gasifier are reactions to produce carbon monoxide (CO), hydrogen ( $\text{H}_2$ ), and a small amount of methane ( $\text{CH}_4$ ). The chemical reaction in the coal gasification process can be divided into two groups: reactions with solid carbon and reaction between gaseous substances. The main chemical reaction is shown in Table 1. The side chemical reactions may occur in the gasifier but these reactions do not have a noticeable effect on the heat and material balance.

To determine the mechanism of reactions in the gasifier for a wide temperature range, the Gibbs free energy was calculated. It is well-known that among several possible reactions, the most probable reaction is the reaction with the minimum of the Gibbs



**Fig. 1.** Schematic view of an industrial gas furnace with thermochemical recuperation of the exhaust gas heat by coal gasification: a) TCR by coal gasification with steam; b) TCR by coal gasification with steam-flue gas mixture.



**Fig. 2.** The comparison of the heat generation processes for the traditional furnaces (a) and for the furnaces with TCR (b).

**Table 1**  
Possible chemical reactions in the gasifier.

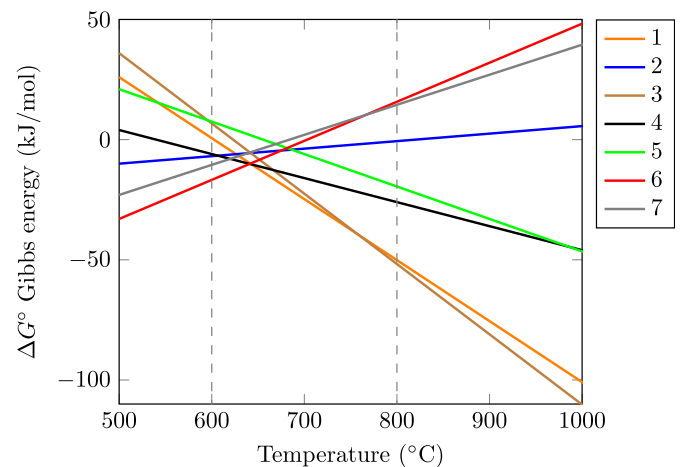
| No.                | Reaction                                    | $\Delta H_{298}$ , (kJ/mol) |
|--------------------|---------------------------------------------|-----------------------------|
| Solid carbon       |                                             |                             |
| 1                  | $C + CO_2 \rightleftharpoons 2CO$           | 173.3                       |
| 2                  | $C + H_2O \rightleftharpoons CO + H_2$      | 90.13                       |
| 3                  | $C + 2H_2O \rightleftharpoons CO_2 + 2H_2$  | 131.3                       |
| 4                  | $C + 2H_2 \rightleftharpoons CH_4$          | -74.8                       |
| Gaseous substances |                                             |                             |
| 5                  | $CO + H_2O \rightleftharpoons CO_2 + H_2$   | -41.1                       |
| 6                  | $CO + 3H_2 \rightleftharpoons CH_4 + H_2O$  | -206.4                      |
| 7                  | $2CO + 2H_2 \rightleftharpoons CH_4 + CO_2$ | -247.3                      |

free energy. In this study, the FactSage database was used to determine the Gibbs free energy change for reactions in Table 1 [33]. The change of the Gibbs free energy as a function of temperature for the chemical reaction from Table 1 is presented in Fig. 3. The number of reaction in Fig. 3 is matching the number of reaction in Table 1.

Fig. 3 shows the reactions of carbon gasification mainly take place in the temperature range above 650 °C. In addition, Fig. 3 depicts that the change in the Gibbs energy for gasification reactions depends significantly on temperature. Therefore, the mechanism of chemical reactions, as well as the endothermic effect of the entire gasification process, depends on temperature. The dependence of the endothermic effect of the gasification process leads to the fact that the efficiency of thermochemical recuperation systems also depends on temperature.

## 2.2. TCR efficiency

To evaluate the effectiveness of the thermochemical recuperation systems, the heat recuperation degree (HRD) was used as an



**Fig. 3.** The change of Gibbs free energy as a function of temperature.

efficiency criterion of such systems. The HRD indicates the ratio of the recovered heat to the heat of the exhaust flue gases. Analyzing the schematic diagrams in Fig. 1, it can be seen that the heat recuperation of the exhaust flue gases is carried out in two elements – a gasifier and a steam generator. Therefore, according to the definition, the heat recuperation degree can be expressed as follows:

$$R = \frac{H_{rec}}{H_{exh}} = \frac{H_{rec}^{gas} + H_{rec}^{st.gen}}{H_{exh}}, \quad (1)$$

where  $H_{rec}$  – a recuperated heat in the TCR system, kJ;  $H_{rec}^{gas}$ ,  $H_{rec}^{st.gen}$  – a recuperated heat in a gasifier and a steam generator,

respectively;  $H_{exh}$  — an exhaust heat, kJ.

The recuperated heat in the gasifier depends on the heat transferred in the gasifier for the gasification reaction and synthesis gas heating. Therefore, the recuperated heat in the gasifier can be determined as follows:

$$H_{rec}^{gas} = H_{reac} + \Delta H_{sg}, \quad (2)$$

where  $H_{reac}$  — enthalpy of the coal gasification process, kJ;  $\Delta H_{sg}$  — increasing enthalpy of synthesis gas in a gasifier due to heating, kJ.

The recuperated heat in the steam generator ( $H_{rec}^{st,gen}$ ) can be determined as follows:

$$H_{rec}^{st,gen} = \Delta Q_{st,gen} = (h_{steam} - c_p \cdot t_w^{in}) G_{steam}^{in}, \quad (3)$$

where  $h_{steam}$  is the enthalpy of steam before a gasifier, kJ/kg;  $G_{steam}^{in}$  is the mass of water for a steam generation, kg;  $t_w^{in}$  is feed water temperature, °C;  $c_p$  is the heat capacity of feed water, kJ/(kg·°C).

To calculate the heat recuperation degree by eq. (1), the composition of the syngas after the gasifier as well as the enthalpy of the gasification process should be determined.

An algorithm of determination of HRD for the TCR systems by coal gasification is presented as a block diagram in Fig. 4. In this algorithm, a steady state operation operating mode was used. Therefore, the heat recuperation degree is independent of the considered time interval.

One of the main advantages of TCR using is increasing in combustion heat of original fuel by a transformation of waste-heat to the chemical energy of new synthetic fuel (syngas). Thus, to assess the degree of such combustion heat increasing, the heat transformation coefficient was determined. According to the definition, the heat transformation coefficient is the ratio of combustion heat of new syngas to the combustion heat of the initial coal. The heat transformation coefficient (HTC) must be written as follows:

$$n_q = LHV_{syngas} / LHV_{coal}. \quad (4)$$

where  $LHV_{syngas}$  — low heating value of syngas produced by gasification of 1 kg of original coal, kJ;  $LHV_{coal}$  — low heating value of 1 kg of original coal, kJ.

Combustion heat of synthetic fuel produced by coal gasification is also depending on composition, and as a consequence of this on

temperature, pressure, and composition of the reaction mixture. Therefore, the heat transformation coefficient also can be determined based on the thermodynamic analysis.

The equilibrium gas compositions of synthesis gas after the coal gasification were calculated via thermodynamic analysis. The thermodynamic analysis based on the Gibbs free energy minimization method was done via software Aspen-HYSYS. A Gibbs reactor in this software is used to determine the composition of synthesis gas at the gasifier outlet in the reaction system. Fig. 5 shows the computational scheme of the coal gasification process with steam (a) and with a mixture of steam and flue gas for Aspen HYSYS.

### 3. Results and discussion

According to Fig. 5, in the thermochemical recuperation systems, the waste heat is used to carry out the gasification reaction and to heat the reaction components. However, synthesis gas consists of combustible components (hydrogen, carbon monoxide) and non-combustible components (unreacted steam, carbon dioxide). An increase in the steam-to-carbon (S/C) ratio leads to an increase in non-combustible components in the synthesis gas. Therefore, in the presented study, the S/C ratios from 0.25 to 2.0 are tested. In addition, the temperature of flue gases after industrial furnaces can vary within wide ranges and depends on the type of industrial furnace [34]. Therefore, the calculations were carried out in a wide temperature range from 200 °C to 1200 °C.

#### 3.1. Verification

The compositions of real coal are quite different. Therefore, for further energy analysis of the thermochemical waste-heat recuperation system by coal gasification the following assumptions were chosen:

- coal is China Datong bituminous coal (LHV = 21236 kJ/kg);
- steam after a steam generator is saturated and pressure is 10 bar;
- temperature of preheated mixture is equal to temperature in the reaction zone.

However, to verify the results of the thermodynamic analysis of

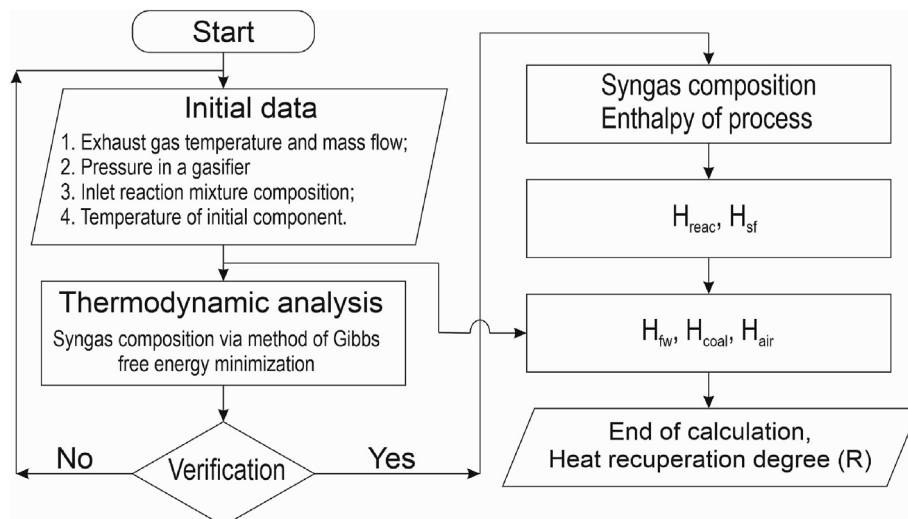


Fig. 4. An algorithm of calculation of a heat recuperation degree for the TCR systems by coal gasification.

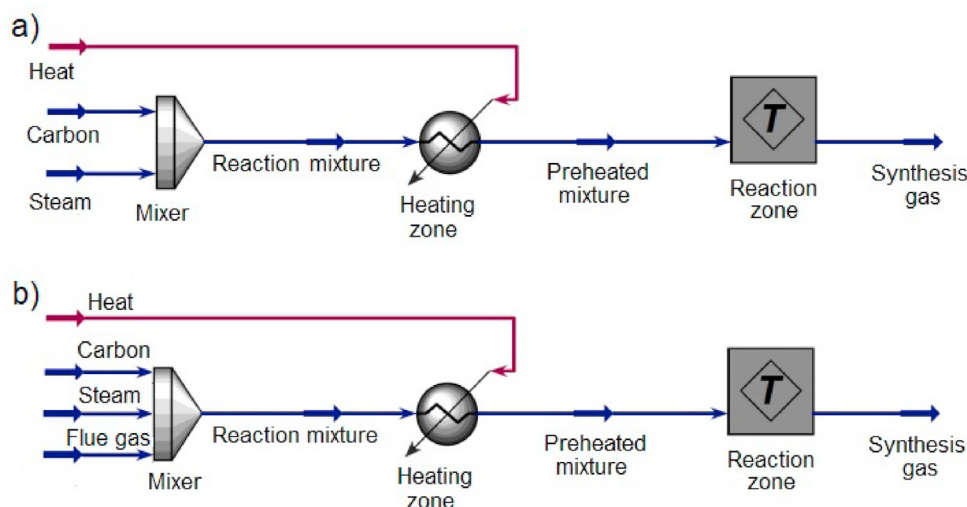


Fig. 5. The computational scheme of the coal gasification process: a) with steam; b) with mixture of steam and flue gas.

the steam gasification process, the results of other authors were used. For verification, the synthesis gas composition obtained after the gasification of China Datong bituminous coal was used [35]. The composition of this coal and other properties were presented in many publications, e.g. Ref. [35]. The composition of China Datong bituminous coal is carbon (64.53%), hydrogen (3.75%), nitrogen (0.95%), sulfur (0.56%), oxygen (7.01%). However, due to the complex composition of the real coal, the coal composition was simplified to three elements: carbon (C), hydrogen ( $H_2$ ), and oxygen ( $O_2$ ). Also, the following assumptions were used for the thermodynamic analysis: no ash in the synthesis gas, no sulfur compounds in the synthesis gas, the coal moisture is zero. The same assumptions were used in other articles dealing with the thermodynamic analysis of the coal gasification process [36,37]. The main goal of this paper is investigation of the thermochemical waste-heat recuperation system by coal gasification. Therefore, the air heating systems and other utility systems were not analyzed because this issue was out of the main goal of the paper.

Fig. 6 shows the comparison of the results of the thermodynamic analysis performed in this paper via Aspen HYSYS with the results obtained by Duan and co-authors [35]. The operation conditions such as the composition of the inlet reaction mixture,

temperature, and pressure were similar in both analyses. An obvious correlation of the results is observed for all components of synthesis gas. There some discrepancies for methane and carbon monoxide mole fraction. To evaluate the discrepancy between the results, the statistical analysis by Pearson criterion was accomplished. The statistical analysis was performed on the basis of the 9-bit histogram and the results of this analysis are presented in Fig. 7.

Fig. 7a shows the normal Gaussian distribution for total mole number and as it can be the good agreement is observed with the law of the normal Gaussian distribution ( $\chi_{func} = 4.09 < \chi_{exp} = 12.592$ , with a confidence level of  $P = 95\%$  and the number of degrees of freedom  $m = 6$ ). Fig. 7b depicts the deviation between results for each analyzed element. The maximum error is observed for methane mole fraction, and the minimum error is for hydrogen and carbon monoxide.

### 3.2. Heat recuperation degree

To calculate the heat recuperation degree, it is necessary to know the enthalpy of the gasification process. The enthalpy of the gasification process shows how much of the heat of exhaust heat has been converted into the chemical energy of synthesis gas.

Fig. 8 shows the dependence of the enthalpy of the gasification process on temperature for various S/C ratios at  $p = 5$  bar. The S/C ratio varied significantly: 4 times less than stoichiometric and 2 times more than stoichiometric. It should be noted that an increase in the S/C ratio above the stoichiometric one can lead to a decrease in the energy efficiency of the TCR systems because the heat input for steam generation increases [38].

HCOALGEN and DCOALIGT models based on proximate and ultimate analyses are used to calculate the enthalpy of gasification. Moreover, the property methods, such as RKSBM, RKSOAVE, or PRBM, are selected when gasification occurs at the temperature range from 200 °C to 1200 °C. Moreover, to check the obtained results for the enthalpy process, the difference in LHV of syngas and LHV of initial mixture. The results were in a good correlation.

Fig. 8 shows that an increase in the S/C ratio leads to an increase in the enthalpy of the process, especially for the S/C ratio below the stoichiometric one. This is due to the fact that the heat balance of the gasification process depends on the material balance. When the S/C ratio increases from 0 to 1, the amount of steam that reacts with coal increases. As a result, the enthalpy of the process increases.

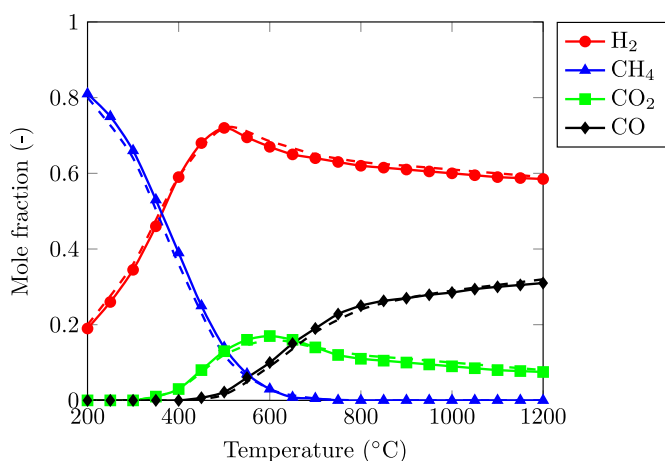


Fig. 6. Comparison of thermodynamic analysis results with results from Ref. [35] obtained at atmospheric pressure and S/C = 2.0.



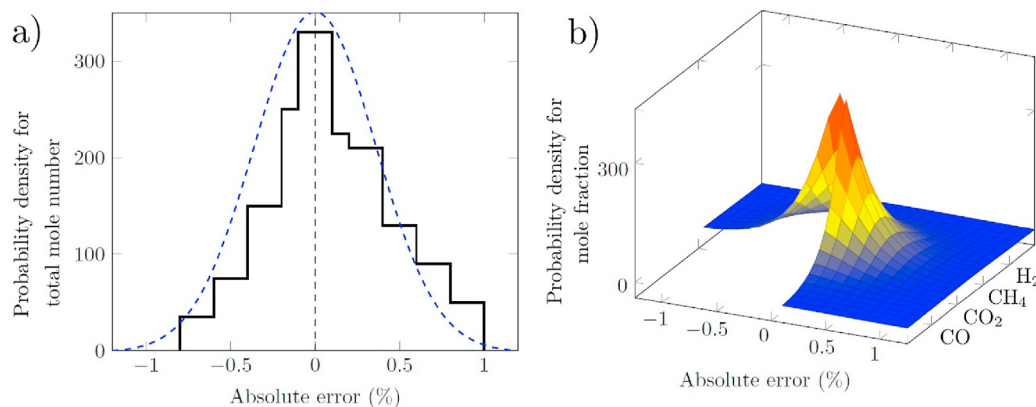


Fig. 7. Comparison of thermodynamic analysis results with results from Ref. [35] obtained at atmospheric pressure and  $S/C = 2.0$ .

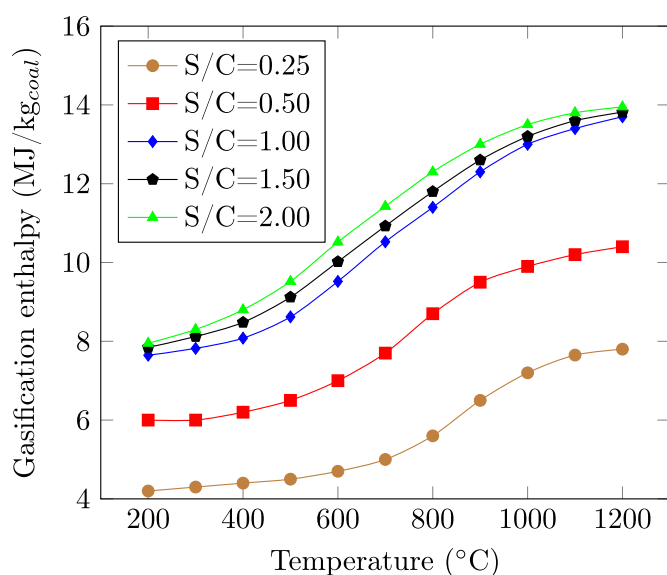


Fig. 8. The enthalpy of the gasification process as a function of temperature for various  $S/C$  ratios at  $p = 5$  bar.

However, with an increase in the  $S/C$  ratio above the stoichiometric value, the increase of process enthalpy significantly decreases. This is due to the fact that all coal is not consumed in chemical reactions, and the remaining coal is not reacted with steam. This steam is ballast. In the TCR systems, an increase in the  $S/C$  ratio above the stoichiometric leads to the fact that the synthesis gas contains remaining steam, for the production of which heat is expended (according to the scheme in Fig. 1a). Therefore, determining the optimal  $S/C$  ratio is an urgent task.

The main criterion for the efficiency of the TCR systems is the heat recuperation degree. Fig. 9 shows the dependence of the heat recuperation degree as a function of temperature at different  $S/C$  ratios at  $p = 5$  bar. The HRD is calculated according to eq. (1). The initial data for the calculation are as follows: the temperature of the feedwater and the temperature of the coal are the same and equal to  $70^\circ\text{C}$ . Exhaust gas consists of combustion products obtained by the combustion of coal with the molecular formula  $\text{C}_{0.696}\text{O}_{0.082}$  (according to the elemental analysis of Datong bituminous coal); steam after a steam generator is saturated and pressure of steam is 5 bar.

Fig. 9 shows that an increase in the  $S/C$  ratio leads to an increase in the HRD. This is due to the fact that the numerator in eq. (1)

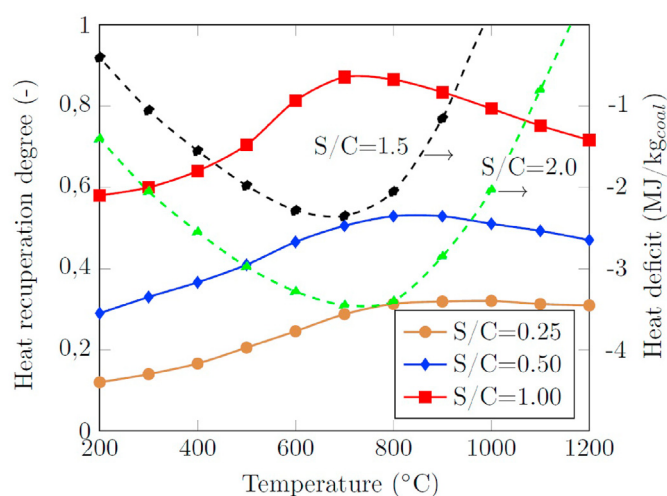


Fig. 9. Heat recuperation degree as a function of temperature for various  $S/C$  ratios at  $p = 5$  bar.

increases. In addition, the numerator in the expression increases not only due to the increase in the enthalpy of gasification but also due to the steam generation. Fig. 9 shows that in the temperature range up to  $700\text{--}800^\circ\text{C}$  HRD increases monotonically and reaches its maximum value at  $720^\circ\text{C}$  for  $S/C = 1.0$ ,  $810^\circ\text{C}$  at  $S/C = 0.5$  and  $930^\circ\text{C}$  at  $S/C = 0.25$ . Further, HRD decreases. This is due to the fact that in eq. (1) the denominator increases (the enthalpy of exhaust gases increases), while the numerator (recuperated heat) remains practically constant.

In addition, Fig. 9 does not show the HRD when  $S/C$  is greater than 1.0. This is due to the fact that in this case, the heat of the exhaust flue gases is not sufficient to generate the steam required for coal gasification. Therefore, in Fig. 9, the dashed line shows the heat deficit for the implementation of TCR due to coal gasification at  $S/C$  greater than 1.0.

One way to avoid a heat deficit in the TCR systems at  $S/C$  greater than 1 is to add flue gases to the reaction mixture. In this case, the coal gasification process in the gasifier will take place in the presence of a mixture of steam and flue gases. Fig. 10 shows the amount of flue gas moles required to eliminate heat deficit in the TCS system. As it can be seen in Fig. 10 the maximum amount of flue gases (about  $20\text{ mol/kg}$  of coal - this corresponds to about  $1\text{ m}^3$  of flue gases at normal conditions) must be added in the temperature range from  $400$  to  $600^\circ\text{C}$  at  $S/C = 1.5$ . For the case  $S/C = 2.0$ , the

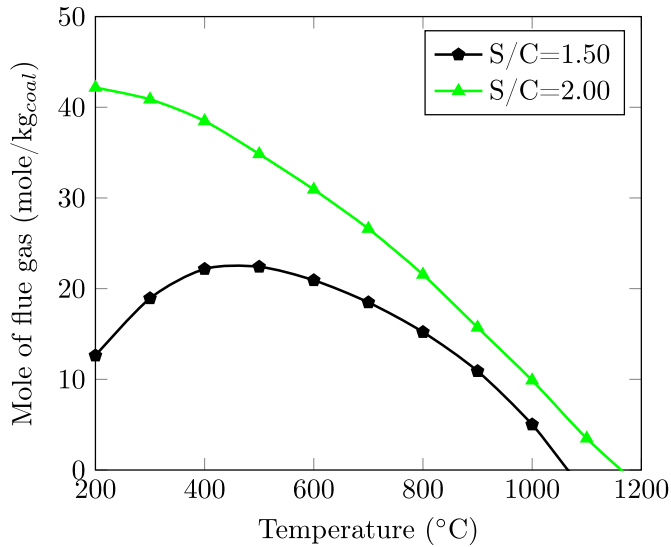


Fig. 10. The number of moles of flue gases to avoid heat deficit in the TCR system by coal gasification with steam-flue gas mixture (Fig.1b).

amount of flue gases decreases monotonically in the temperature range 200–1200 °C.

Thermochemical recuperation by coal steam gasification has a distinctive feature in comparison with the thermochemical recuperation by steam reforming of natural gas. Gasification of 1 kg of coal requires approximately 1.5 kg of steam. Therefore, an increase in the S/C ratio in the reaction mixture leads to additional heat consumption for steam production. The effect of the S/C ratio on HRD is shown in Fig. 11. In Fig. 11, the solid lines show HRD calculated according to eq. (1). For S/C > 1, the heat deficit is taken into account in the numerator, according to the eq. (5). This HRD can be called gross HRD.

$$R_{gross} = \frac{H_{rec}}{H_{exh}} = \frac{H_{rec}^{gas} + H_{rec}^{st.gen} - H_{def}}{H_{exh}}, \quad (5)$$

where  $H_{def}$  – a heat deficit in the TCR system, kJ.

At S/C > 1, the gasification products (synthesis gas) contain residual steam. In this case, this steam in synthesis gas does not participate in the combustion process because this is a non-combustion component. Therefore, the enthalpy of the residual

steam in the synthesis gas cannot be considered in its pure form as recuperated heat. Residual steam is ballast. To take this fact into account, I introduced the concept of net HRD, which can be calculated according to the following expression:

$$R_{net} = \frac{H_{rec}}{H_{exh}} = \frac{H_{rec}^{gas} + H_{rec}^{st.gen} - H_{def} - H_{H_2O}^{res}}{H_{exh}}, \quad (6)$$

where  $H_{H_2O}^{res}$  – residual steam in synthesis gas after a gasifier, kJ.

Fig. 11 shows that the maximum net HRD takes place at a temperature of 800 °C and S/C = 1. In addition, an increase in S/C up to 1.0 leads to an increase in both net HRD and gross HRD. However, a further increase in S/C leads to a decrease in both net HRD and gross HRD because, firstly, there is a heat deficit, and secondly, residual steam has a negative effect on the heat balance of the TCR system. However, it should be noted that for temperatures above 1000 °C, the gross HDR increases with an increase in the S/C ratio > 1, since in this case, there is no heat deficit in the TCR system (according to Fig. 9). It can be concluded from Fig. 11 that the theoretical maximum net HRD can be achieved at S/C = 1 at a temperature of 800 °C.

### 3.3. Heat transformation coefficient

The basic concept of thermochemical recuperation is the transformation of waste-heat into the chemical energy of new synthetic fuel. This transformation is an increase in the low heating value of the new fuel compared to the original fuel. The heat transformation ratio shows the ratio of the heat of combustion of the new synthetic fuel to the heat of combustion of the original fuel according to expression (4).

Fig. 12a shows the dependence of the heat transformation coefficient on the gasification temperature. An increase in the S/C ratio leads to an increase in the heat transformation coefficient. With an increase in the S/C ratio above the stoichiometric ratio, HTC increases slightly. Despite the fact that an increase in the S/C ratio leads to an increase in HTC as well as to an increase in the total heat of combustion, the specific low heating value of synthesis gas (per 1 kg of syngas) decreases. Fig. 12b shows the dependence of the specific low heating value of 1 kg of synthesis gas. Fig. 12b shows that the addition of steam in the initial reaction mixture significantly reduces the specific LHV of synthesis gas.

### 3.4. Heat balance

To compile the heat balance of the TCR system that shown in Fig. 1a, the following initial data were taken:

- The composition of flue gas (mole fraction of each element) after a furnace:  $CO_2 = 0.14$ ;  $H_2O = 0.08$ ;  $N_2 = 0.75$ ;  $O_2 = 0.03$ ;
- The temperature of weed water is 70 °C, the temperature of coal before the mixer-exchanger (Fig. 5) is 20 °C;
- The temperature of steam after the steam generator is 200 °C ( $p = 5$  bar);
- The gasification temperature is 200 °C less than the temperature of flue gas after the furnace;
- The synfuel temperature is equal to the reforming temperature;
- The steam-to-carbon ratio is 1. All calculations are performed for 1 kg of coal.

The following notation is used for the heat inputs and outputs. FG is the stream of exhaust flue gases after the furnace; FG' is the stream of exhaust flue gases after the gasifier; Steam is steam; Mixture is steam-coal mixture; SF is synthetic fuel (synfuel);  $H_{gasif}$  is

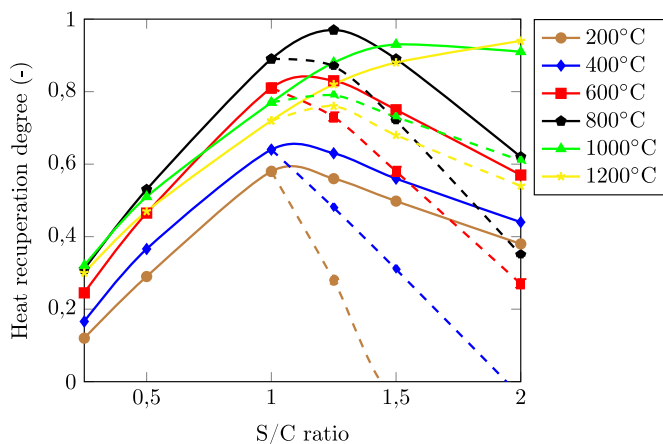


Fig. 11. Effect of the S/C ratio on heat recuperation degree for various temperatures: solid lines – gross HRD; dashed lines – net HRD.

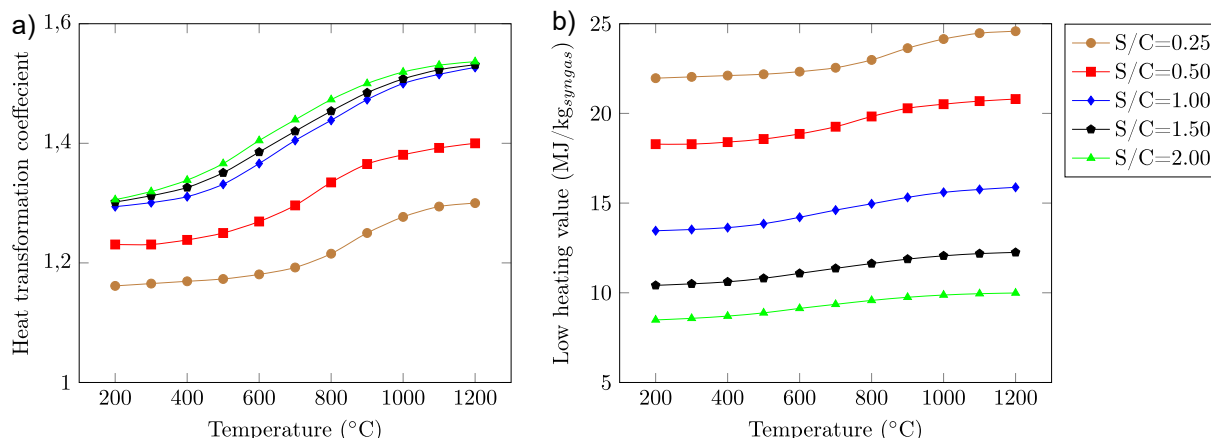


Fig. 12. a) Heat transformation coefficient vs. temperature for various S/C ratios; b) Specific low heating value of 1 kg of synthesis gas vs. temperature for various S/C ratios.

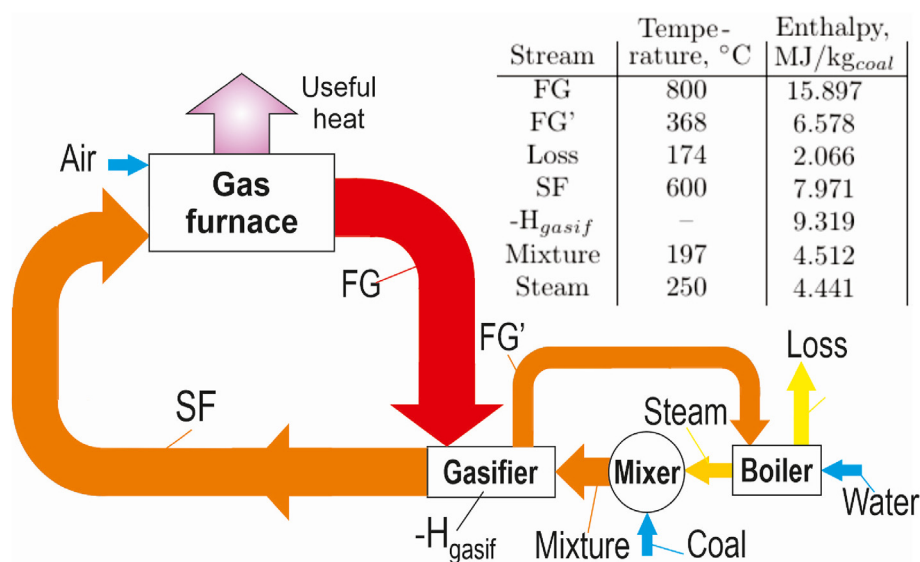


Fig. 13. The heat balance of the TCR system when the temperature of exhaust gas is 800 °C and steam-to-carbon ratio is 1.0.

enthalpy of the gasification reaction; *Loss* is the flue gases after the steam generator.

Fig. 13 shows the heat balance of the TCR system when the temperature of exhaust gas is 800 °C and steam-to-carbon ratio is 1.0.

#### 4. Conclusion

The efficiency of the thermochemical waste-heat recuperation (TCR) systems by coal gasification were determined. As the main criterion of TCR efficiency a heat recuperation degree (HRD) was introduced. Moreover, HRD was split to gross and net HRD. It was determined that in the temperature range up to 700–800 °C HRD increases monotonically and reaches its maximum value at 720 °C for S/C = 1.0, 810 °C at S/C = 0.5 and 930 °C at S/C = 0.25. The maximum net HRD takes place at a temperature of 800 °C and S/C = 1. In addition, an increase in S/C up to 1.0 leads to an increase in both net HRD and gross HRD. For S/C above 1.0, the heat of the exhaust flue gases is not sufficient to generate the steam required

for coal gasification because there is a heat deficit. For temperatures above 1000 °C, the gross HDR increases with an increase in the S/C ratio > 1, since in this case, there is no heat deficit in the TCR system. The theoretical maximum net HRD of 0.97 can be achieved at S/C = 1 at a temperature of 800 °C. The heat transformation coefficient (HTC) was also considered as a criterion of efficiency of the TCR systems. The maximum HTC is above 1.5 for S/C greater than 1.0. The thermochemical recuperation systems with coal gasification will have a maximum efficiency for the industrial with temperature of exhaust gases from 700 °C to 1000 °C. However, the detailed techno-economic assessment has to be performed for exact determination of the furnace efficiency with TCR.

The temperature distribution at the air heater system and other utility systems strongly depends on the type of furnace. The main goal of this paper is investigation of the thermochemical waste-heat recuperation system by coal gasification. However the temperature distribution in such systems will be analysed in the author's next work, where a comparison of a forging furnace with TCR and an air heater will be made.



## Author statement

This paper was written by Dmitry Pashchenko as a sole author.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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